

Customer Shopping Behavior Analysis

A comprehensive data analytics project exploring revenue drivers, seasonal trends, loyalty patterns, and discount effectiveness across 3,900 transactional records.

MYSQL

PYTHON

POWER BI

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Project Objectives

Demographics

Identify which customer demographics generate the most revenue

Seasonal Trends

Uncover seasonal purchase behavior by product category

Discount Impact

Measure the effect of discounts on purchase amounts

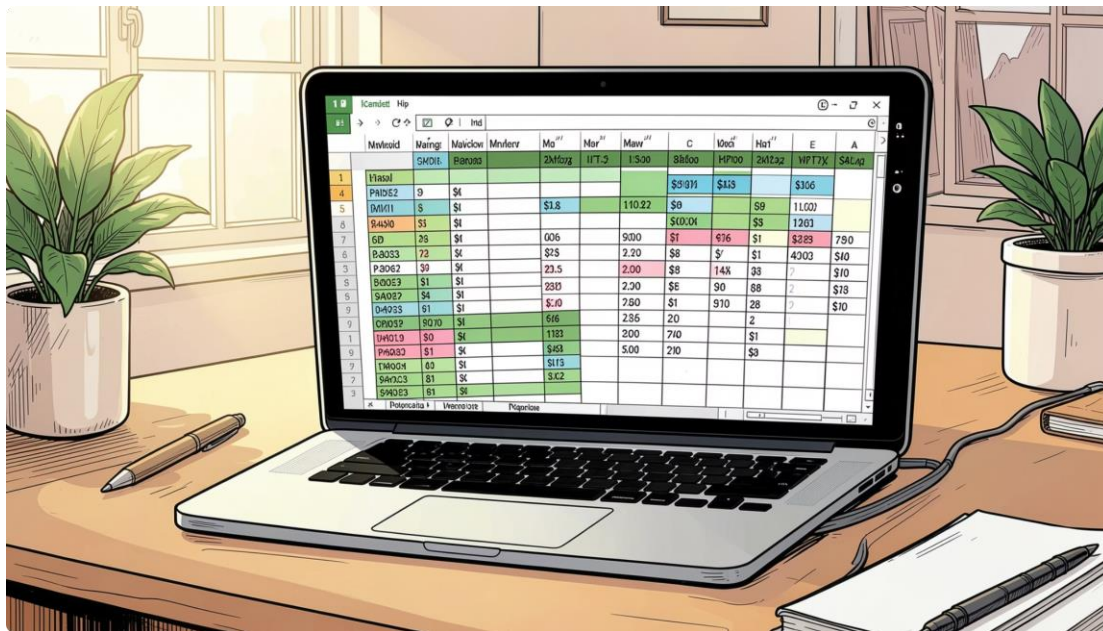
Customer Segments

Segment customers based on purchasing patterns and average spend

Shipping & Value

Understand the relationship between shipping preference and order value

Dataset Summary



Dataset at a Glance

3,900 observations across **18 feature dimensions** spanning three key categories:

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item Purchased, Category, Purchase Amount, Size, Color, Season

Shopping Behavior

Discount Applied, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping Type

Exploratory Data Analysis — Python

Initial workflow used **pandas** for structural assessment and statistical profiling.

01

Data Loading & Exploration

Imported dataset; used `df.info()` and `df.describe()` for structure and summary stats.

03

Standardization & Feature Engineering

Renamed columns to `snake_case`; created `age_group` and `purchase_frequency_days` columns.

02

Missing Data Handling

Checked for nulls; imputed missing Review Ratings using the median per product category.

04

Data Consistency & Database Integration

Dropped redundant `promo_code_used`; loaded cleaned DataFrame into PostgreSQL for SQL analysis.

Summary Statistics

3,900

Total Records

Unique transactional observations

44

Avg. Age (yrs)

Age range: 18–70 years

\$59.76

Avg. Purchase

Range: \$20–\$100, Std Dev: \$23.69

2,652

Male Customers

Largest gender segment in dataset

1,737

Clothing Purchases

Most purchased product category

Data Analysis — SQL

Structured PostgreSQL queries answered eight key business questions:

1

Revenue by Gender

Compared total revenue generated by male vs. female customers

2

Revenue by Age Group

Identified which age segments contribute most to revenue

3

Top Product Categories

Ranked categories by average review ratings

4

Subscription vs. Non-Subscription

Compared average spend between subscriber and non-subscriber groups

5

Seasonal Trends

Analyzed purchase patterns across seasons by category

6

Shipping Preference

Examined which shipping types drive higher average revenue

7

Customer Segmentation

Classified customers as New, Returning, or Loyal by purchase frequency

8

Discount Impact

Measured how discounts affect average purchase amounts

Key Findings

Age Group Revenue

Mid-age customers generate disproportionately higher revenue — **\$110,945** — indicating prime target demographics for marketing

Seasonal Patterns

Clothing peaks in Spring (**\$27,692**); Accessories and Outerwear peak in Fall (**\$19,847 and \$5,259**), footwear in spring (**\$9555**)

Discount Effect

Discounts influence purchase amounts, requiring careful balance between volume growth and margin control.

Customer Segments

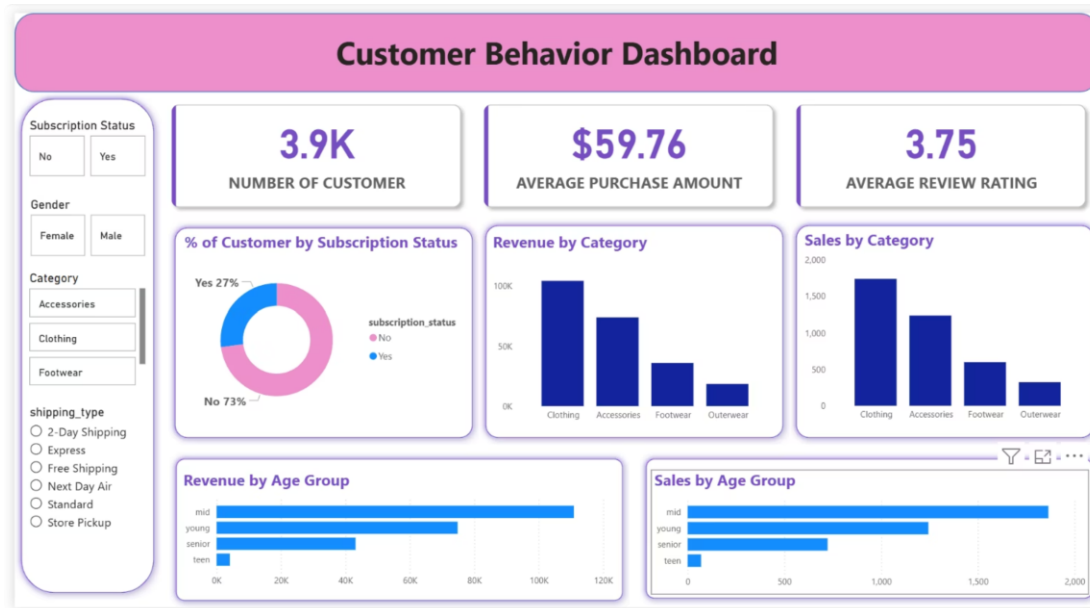
Majority of customers fall in the '**Loyal**' segment with avg spend of **\$59.50**.

Shipping Preference

2-Day Shipping users tend to have higher average order values.



Power BI Dashboard



Dashboard KPIs

- **73%** Non-Subscribers vs. **27%** Subscribers
- Clothing generated the highest revenue and sales volume
- Mid-age customers contributed the highest revenue and sales volume

3.9K

Total Customers

\$59.76

Avg. Purchase Amount

3.75

Avg. Review Rating

Business Recommendations



Boost Subscription Growth

Design and promote exclusive, high-value benefits to increase the subscriber base and drive higher per-customer revenue



Refine Loyalty Frameworks

Reward repeat buyers to move them into the 'Loyal' segment



Review Discount Policy

Balance sales boosts with margin control by targeting discounts strategically



Optimize Product Positioning

Aggressively highlight top-rated items and best-sellers (specifically within the "Clothing" category) in digital storefronts to leverage social proof



Execute Targeted Marketing

Focus advertising spend on top-performing age group cohorts and prioritize retention efforts toward users who frequently select express shipping, as they represent a high-value segment

Project Summary



This end-to-end pipeline transformed raw transactional data into actionable business intelligence, supporting data-driven decisions across marketing, loyalty, and pricing strategy.

Tools	Data	Key Insights
MySQL · Python · Power BI	3,900 records · 18 features	5 actionable recommendations